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Inventor(s)	Vanhee; Steven et al.

Electric machine with a cooling assembly

Abstract

Methods and systems for an electric machine. The electric machine includes, in one example, a stator with end windings and a stator core and a cooling assembly with a spray ring that is directly coupled to an axial face of the stator core. In the electric machine, the spray ring is configured to spray a coolant at the end windings and includes a first section that extends down the axial face of the stator core and a second section that extends axially away from the stator core.

Inventors:	Vanhee; Steven (Hooglede, BE), Druant; Joachim (Houthulst, BE)
Applicant:	Dana Automotive Systems Group, LLC (Maumee, OH)
Family ID:	91078869
Assignee:	DANA AUTOMOTIVE SYSTEMS GROUP, LLC (Maumee, OH)
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Primary Examiner: Johnson; Eric
Attorney, Agent or Firm: McCoy Russell LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to an electric machine with spray rings that direct coolant through stator end windings.

BACKGROUND AND SUMMARY

(2) In electric motors, such as electric motors in electric vehicle (EV) applications, cooling systems has been used in an attempt to achieve greater motor efficiency. For instance, oil may be directed onto stator end windings to increase motor efficiency. However, the flow pattern in the end winding cavity may not achieve a targeted amount of end winding cooling in certain motor designs.

(3) US 2021/0044172 A1 Hombsch et al. teaches an electric motor with a cooling jacket that circulates oil around an outer surface of a stator and an inner surface of the cooling jacket. Hombsch's motor further includes a ring that is coupled to the stator via a seal and an orifice that attempts to direct oil at the end windings.

(4) The inventors have recognized several issues with Hombsch's motor as well as other previous motor. For instance, the use of the seal to connect the spray ring to the stator in Hombsch's motor, decreases the compactness of the motor and increases the complexity of the motor which can lead to a lengthy assembly process. Previous spray rings have been constructed from aluminum. When aluminum spray rings are used, creepage distances should be respected, thereby constraining the

location of the jet.

(5) To overcome at least some of the abovementioned issues the inventors developed an electric machine. The electric machine includes, in one example, a stator with end windings and a stator core. The electric machine further includes a cooling assembly with a spray ring that is directly coupled (e.g., bonded) to an axial face of the stator core. In such an example, the spray ring sprays a coolant at the end windings. The spray ring includes a first section that extends down the axial face of the stator core and a second section that extends axially away from the stator core. Shaping the spray ring in this manner enables a greater amount of coolant to flow through the end windings, thereby increasing stator cooling and machine efficiency as a consequence. Further, shaping the spray ring in this manner, significantly simplifies the assembly process. Further, the spray ring enhances sealing with the stator face due to the large planer contact area between the spray ring and the stator core.

(6) Further, in one example, the stator end windings may be hairpin end windings. The use of the hairpin end windings in conjunction with the L-shaped spray ring allows the compactness and efficiency of the machine to be increased, if desired. To elaborate, the hairpin end windings allows the power density of the machine to be increased, in comparison to a machine with round end windings, for instance.

(7) Still further, in one example, the spray ring may be coupled to a stack of laminations in the stator using bonding and impregnation of the ring onto the lamination stack. Consequently, the space efficiency of the end winding assembly may be further increased and the longevity of the machine may be increased, when compared to machines which attach a spray ring to the stator via a seal.

(8) Further, in one example, the spring ring may include a third section that extends axially away from the stator core. In such an example, the third section may be positioned at an inner diameter of the spray ring. The third section of the spray ring may shield coolant spray from the rotational airflow originating from the rotor and, therefore, increases the effectiveness of the cooling methodology.

(9) It should be understood that the summary above is provided to introduce in a simplified form a selection of concepts that are further described in the detailed description. It is not meant to identify key or essential features of the claimed subject matter, the scope of which is defined uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not limited to implementations that solve any disadvantages noted above or in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 shows a cross-sectional view of an example of an electric machine with a cooling assembly.

(2) FIGS. 2A-2B show detailed front and rear views of an example of a spray ring in an electric machine cooling assembly.

(3) FIG. 3 shows a detailed view of an example of a stator end winding cooling assembly.

(4) FIG. 4 shows a cross-sectional view of another example of an electric machine with a cooling assembly.

(5) FIG. 5 shows a detailed view of a section of the electric machine, depicted in FIG. 4.

(6) FIG. 6 shows another example of a spray ring in a cooling assembly for an electric machine.

DETAILED DESCRIPTION

(7) An electric machine, is described herein, that includes a cooling assembly that effectively cools stator end windings to a greater extent than previous motor cooling systems. To achieve the end

winding cooling, the electric machine includes a spray ring that has an “L” or “C” type shape which generates a coolant flow in the end winding cavity that directs a greater portion of the coolant directly through the end windings to remove more heat from the windings when compared to other cooling systems. As a result, machine efficiency is increased, thereby increasing machine appeal. Still further, the end windings may specifically be hairpin style end windings to enable the power density of the machine to be increased, if desired. The spray ring may further include a section at an inner diameter that axially extends outward from the stator's core. This spray ring section shields coolant spray from the rotational airflow originating from the rotor and, therefore, increases the effectiveness of the cooling methodology.

(8) FIG. 1 shows an example of an electric machine with a cooling assembly that includes an L-shaped spray ring that strategically directs a greater amount of coolant through the stator's end windings than other machine cooling systems. FIGS. 2A-2B show front and rear views of an example of the L-shaped spray ring that alter the coolant flow dynamics in the end winding cavity to increase stator cooling. FIG. 3 shows a detailed illustration of another example of stator assembly in an electric machine which increases the amount of heat removed from the end windings, thereby increasing motor efficiency. FIG. 4 shows another example of an electric machine with a cooling assembly. FIG. 5 shows a detailed view a spray ring in the cooling assembly, depicted in FIG. 4. FIG. 6 shows another example of a spray ring for an electric machine cooling assembly.

(9) FIG. 1 shows an illustration of an electric machine **100**. The electric machine **100** may be designed as an electric motor-generator and may be included in a system **102** which may take a variety forms. For instance, the electric machine **100** may be incorporated into an electric drive of an electric vehicle (EV). In the EV example, the EV may be an all-electric vehicle (e.g., a battery electric vehicle (BEV)), in one example, or a hybrid electric vehicle (HEV), in another example. For instance, the electric machine may be mechanically coupled to a transmission (e.g., gearbox) that is coupled to drive wheels using one or more differentials for example. Further, in the EV example, the electric machine may be a traction motor that delivers mechanical power to drive wheels. In the HEV example, the electric machine may be included in an electric axle, and an internal combustion engine may provide motive power to another drive axle. However, the motor may be used in other suitable systems (e.g., stationary systems), in other examples, such as in industrial machines, agricultural systems, mining systems, and the like.

(10) The electric machine **100** includes a rotor **104** that electromagnetically interacts with a stator **106** to drive rotation of a rotor shaft **108** that is included in the rotor. The stator **106** at least partially surrounds the rotor **104**. As such, the electric machine **100** may be a radial flux style motor.

(11) The electric machine **100** in the illustrated example includes a housing **110**. The housing **110** may form an enclosure (e.g., a sealed enclosure) **112** that contains stator end windings **114**, cooling assembly components, and the like, which are expanded upon herein. The housing may include an electrical interface. The electrical interface may be a multi-phase electrical interface with multiple electrical connectors. The electrical interface may be three-phase interface in one example or a six or nine phase interface, in other examples. More generally, the electric machine **100** may be a multi-phase alternating current (AC) machine. However, in other examples, the electric machine **100** may be a direct current (DC) machine.

(12) As illustrated in FIG. 1, the electric machine **100** may be electrically coupled to an inverter **118**. The inverter **118** is designed to convert direct current (DC) power to alternating current (AC) power and vice versa. As such, the electric machine **100** may be an AC electric machine, as indicated above. However, in other examples, the electric machine **100** may be a DC electric machine (as previously indicated), and the inverter **118** may therefore be omitted from the system **102**. The inverter **118** may receive electric energy from one or more energy storage device(s) **120** (e.g., traction batteries, capacitors, combinations thereof, and the like). Arrows **122** signify the

electric energy transfer between the electric machine **100**, the inverter **118**, and the energy storage device(s) **118** that may occur during different modes of system operation.

(13) The rotor **104** may include a core **124** that includes a stack of laminations **125**. In the stack, laminations that may be formed of a metal (e.g., steel, aluminum, combinations thereof, and the like) are sequentially arranged and coupled (e.g., bonded and/or welded) to one another. Further, the rotor lamination stack may include teeth.

(14) Continuing with FIG. **1**, the stator **106** may include a core **126** through which windings extend. These windings protrude from the stator core on either axial end to form the end windings **114**. The end windings may be positioned on opposing axial sides **128** and **130** of the electric machine.

(15) The stator core **126** may include a stack of laminations **131** which may include teeth and gaps at an inner periphery. It will be understood that the stack of stator laminations may have openings through which the windings extends. The stator lamination stack includes multiple metal (e.g., steel, aluminum, combinations thereof, and the like) sheets that are sequentially arranged and may be stacked, welded, or bonded together, in different instances.

(16) The stator core **126** and end windings **114** are schematically illustrated in FIG. **1**. However, it will be understood that they have greater structural complexity that is expanded upon herein. The end windings **114** may be hairpin end windings, in one example, which allows the power density of the motor to be increased, when compared to other types of end windings such as end windings with round cross-sections. A hairpin end winding is defined herein as an end winding with wires that have a rectangular cross-section which are referred to as flat wires. In other examples, end windings with round cross-sections may be used in the electric machine **100** which may however decrease the machine's power density.

(17) The end windings **114** generate heat during machine use. As such, cooling of the end windings to increase motor efficiency may be desired. A cooling assembly **132** is provided to remove heat from the end windings **114**.

(18) The cooling assembly **132** in the illustrated example, includes a coolant pump **134** and a filter **136** that includes a pick-up **138** in a sump **140**. The sump **140** is positioned in an internal enclosure of the electric machine **100** and is contoured to collect coolant **142** that is sprayed at the end windings **114**. The coolant used in the cooling assembly **132** may specifically be oil (e.g., natural and/or synthetic oil), in one example. However, other suitable types of coolant have been contemplated. The pump **134** and filter **136** are incorporated into the electric machine **100** in the illustrated example. However, in other examples, the pump and/or filter may be spaced away from the machine's housing or externally coupled thereto. The location of the pump may be selected based on the machine's end-use packaging constraints, for instance.

(19) To direct coolant towards the end windings, the cooling assembly **132** may include coolant conduits **144**. The coolant conduits **144** may extend through the stator **106** and specifically the stator core **126**. Additionally or alternatively, the coolant conduits **144** may extend through the housing **110**. To elaborate, in other cooling assembly designs, the coolant may be routed through the housing **110** into cavities **146** in the housing **110**.

(20) Further, the coolant conduits **144** may each open into one of the cavities **146** that is arranged between an associated spray ring **148** and the housing **110**. However, alternative coolant routing schemes may be used in other examples. For instance, the coolant conduits may include outlets that are in direct fluidic communication with coolant openings **150** in the spray rings **148**.

(21) In the illustrated example, the housing **110** includes recesses **152** in which a portion of the spray rings **148** resides. In this way, the spray rings **148** may be effectively incorporated into the housing **110**. However, other housing profiles have been contemplated.

(22) The electric machine **100** includes multiple spray rings **148**. To elaborate, in the illustrated example, the electric machine **100** includes a pair of spray rings, the first spray ring positioned on the first axial side **128** of the machine and the second spray ring positioned on the second axial side

130 of the machine. The spray rings **148** may have an annular shape to enable the rotor shaft **108** to pass therethrough.

(23) The spray rings **148** that are directly coupled to the stator **106**. To elaborate, the spray rings **148** are each coupled to an axial face **154** of the stator core **126**. For instance, the spray rings **148** may each be bonded and/or impregnated to an outer lamination in the stack of stator laminations **131**. For instance, the spray rings may be adhesively bonded to the outer lamination in the stack.

(24) Each of the spray rings **148** include a first section **156** that extends down the axial face **154** of the stator **106** (e.g., the stack of stator laminations **131** in the stator core **126**) are is coupled thereto. Each of the spray rings **148** further includes a second section **157** that extend axially away from the stator core **126**. Thus, the profile of the spray rings in radial cross-section have is L-shaped. Designing the spray rings with L-shaped profiles enables a coolant flow pattern to be achieved in which a greater amount of heat is extracted from the end windings.

(25) The spray rings **148** each may further include the coolant opening **150** (e.g., spray channels). The openings **150** receive coolant from the cavities **146** in the housing **110** that is in fluidic communication with the coolant conduits **144**. However, as previously indicated, the coolant conduits **144** may be in direct fluidic communication with the openings **150**.

(26) The system **102** may additionally include a control sub-system **180** with a controller **182**. The controller **182** includes a processor **184** and memory **186**. The memory **186** may hold instructions stored therein that when executed by the processor **184** cause the controller **182** to perform the various methods, control techniques, and the like, described herein. The processor **184** may include a microprocessor unit and/or other types of circuits. The memory **186** may include known data storage mediums such as random access memory, read-only memory, keep alive memory, combinations thereof, and the like.

(27) The controller **182** may receive various signals from sensors **188** positioned in different locations in the system **102**. The sensors **188** may include an electric machine speed sensor, energy storage device temperature sensor(s), an energy storage device state of charge sensor(s), an inverter power sensor, and the like. The controller **182** may also send control signals to various actuators **190** coupled at different locations in the system **102**. For instance, the controller may send signals to the inverter **118** to adjust the rotational speed of the electric machine **100**. In another example, the controller **182** may send a command signal to the electric machine **100** and/or the inverter **118** and in response motor speed may be adjusted. The other controllable components in the system **102** may function in a similar manner with regard to command signals and actuator adjustment.

(28) The system **102** may also include one or more input device(s) **192** (e.g., an accelerator pedal, a brake pedal, a console instrument panel, a touch interface, a touch panel, a keyboard, combinations thereof, and the like). The input device(s) **192**, responsive to user input, may generate a motor speed adjustment request.

(29) An axis system is provided in FIG. 1, as well as FIGS. 2A-6, for reference. The z-axis may be a vertical axis (e.g., parallel to a gravitational axis), the x-axis may be a lateral axis (e.g., horizontal axis), and/or the y-axis may be a longitudinal axis, in one example. However, the axes may have other orientations, in other examples. Rotational axes **199** of the electric machine **100** is further provided for reference in FIG. 1 as well as FIGS. 2A-2B and 4. Cutting planes for the cross-sectional views shown in FIGS. 1, 3, and 4-6 extend through the rotational axes **199**.

(30) FIGS. 2A-2B show respectively front and rear views of an example of a spray ring **200**. The spray ring **200** may be an example of the spray rings **148** included in the electric machine **100** shown in FIG. 1 and/or the other electric machines described herein. The spray ring **200** may have an annular shape with a central opening **202** which allows the rotor shaft to extend therethrough.

(31) FIG. 2A shows a first section **204** and a second section **206** of the spray ring **200** on an outer side **208**. The outer side of the spray ring may be in the enclosure in which the end windings are positioned, when the ring is installed in the machine. When assembled the first section **204** extends down an axial face of the stator core and the second section **206** extends axially away from the

stator core (which is in the direction out of the page in the frame of reference depicted in FIG. 2A), as indicated above. The first and second sections **204** and **206** of the spray ring may form a continuous structure (e.g., a monolithic structure), in one example. For instance, the first and second sections of the spray ring may be manufactured by machining a single piece of material. For instance, the spray ring may be injection molded or manufacturing using other suitable techniques. However, in other examples, the sections of the spray ring may be separately manufactured and then attached to one another via adhesive bonding, interference fitting, combinations thereof, and the like. It will be appreciated that the first section **204** may include openings through which the end windings pass.

(32) FIG. 2B shows the rear side **210** of the spray ring **200** with a surface **212** that may be in face sharing contact with the axial face of the stator core when the spray ring is installed in the electric machine. To elaborate, the surface **212** of the spray ring may be bonded (e.g., bonded and impregnated) onto an outer lamination in the stack of stator laminations of the stator core. In this way, the space efficiency of the machine may be increased.

(33) FIG. 3 shows an example of an electric machine **300**. The electric machine **300** shown in FIG. 3 as well as the other machines described herein may have at least a portion of the structural and/or functional features of the electric machine **100** shown in FIG. 1 and vice versa.

(34) The electric machine **300** includes a housing **302**, a stator **304**, and a rotor **306**. An air gap **308** is formed between the stator **304** and the rotor **306**. The stator **304** includes a core **310** and end windings **312** that extend axially outward from the core. The end windings **114** are depicted as hairpin type end windings in the illustrated embodiment. However, it will be understood, that the end windings may have a round cross-section, as previously indicated.

(35) The end windings **312** may be demarcated into different sections. For instance, the end windings **312** include tips **314** and intermediary sections **316**. The tips **314** may extend in an axial direction while the intermediary sections may extend in both axial and radial directions. However, other contours of the end winding sections may be used in other examples.

(36) The housing **302** is shown including a recess **318** shaped and sized to receive a portion of a section **320** of a spray ring **322**. In this way, the compactness of the machine may be increased. However, other housing profiles may be used, in other embodiments. The housing **302** further forms a compartment **324** that may receive coolant from a cooling assembly, such as the cooling assembly **132** depicted in FIG. 1. The section **320** of the spray ring **322** extends axially away from the stator core **310**. The section **320** may include an interior surface **326** that is stepped to generate a coolant flow pattern that is conducive to increasing stator end winding cooling. A coolant cavity **327** is formed radially inward from the interior surface **326**. Coolant in the cavity **327** is directed through the end windings **312** to increase heat removal therefrom.

(37) Another section **328** of the spray ring **322** extends down the axial face **330** of the stator core **310**. The sections **322** and **328** of the spray ring **322** form an angle **329** with one another. The angle may be approximately 90° in one example. However, more generally the angle **329** may be less than or equal to 90°, in certain embodiments.

(38) Specifically, an inner surface **331** of the spray ring section **328** may be in face sharing contact with the axial face **330** of the core **310**. To elaborate, the surface **331** of the spray ring section **328** may be bonded and impregnated (e.g., adhesively bonded and impregnated) to the axial face **330** of the stator core. In the illustrated example, a seal (e.g., an O-ring seal) between the spray ring and the stator core end face is omitted to increase machine compactness. However, a seal may be used to couple the spray ring to the stator, in other examples, which may come at the expense of machine compactness.

(39) The section **328** of the spray ring **322** allows the electric machine **300** to achieve a desired amount of insulative properties with regard to the stator core **310**, while also increasing the amount of coolant that is directed between the end windings **312**. To elaborate, the section **328** insulates the end windings **312** from the stator iron.

(40) The housing **302** may include extensions **340** that function as heat fins to further increase electric machine cooling and consequently increase machine efficiency. However, in other examples, the extensions may be omitted from the machine.

(41) In one example, the electric machine may forego a water jacket that surrounds the housing due to the level of end winding cooling achieved by the spray ring arrangement. However, the electric machine **300** may include a water jacket that at least partially surrounds the housing and circulates coolant therethrough, in other examples. In such an example, the coolant may be a mixture of water and glycol or oil, for instance.

(42) A balancing plate **342** may be coupled to the rotor **306** via a sealing interface **344**. The balancing plate **342** allows rotational imbalances in the rotor to be reduced. As a result, the likelihood of machine degradation is reduced.

(43) FIG. 4 shows another example of an electric machine **400**. The electric machine **400** again includes a cooling assembly **402** with a pump **404** and coolant conduits **406** (e.g., coolant lines) that direct coolant into selected areas of the machine. In the illustrated example, the coolant conduits **406** route coolant through a housing **408** directly into coolant compartments **410** which are adjacent to spray rings **412**. To elaborate, the coolant compartments **410** may be positioned adjacent to an outer diameter of the spray rings. The spray rings **412** are positioned on opposing axial sides the machine and are directly coupled to (e.g., bonded to) a stator core **414** of a stator **416**. Thus, in the cooling assembly architecture depicted in FIG. 4, coolant is not routed through the stator core. However, as previously indicated cooling assemblies with stator core coolant routing has been contemplated.

(44) The spray rings **412** again each include a first section **418** that extends down an axial face **420** of the stator core **414** and a second section **422** that extends axially away from the stator core. The second section **422** is positioned at or near an outer diameter of the spray ring. However, in the spray ring embodiment shown in FIG. 4, the spray rings **412** each include a third section **424** that extends axially away from the stator core at or near an inner diameter **426** of the spray ring.

(45) A spray channels **428** may extend (e.g., radially extend) through the second sections **422** of the spray ring **412**. The spray channels **428** direct coolant spray towards the end windings **430** of the stator **416** to remove heat therefrom and enhance stator cooling. A cavity **432** formed in the housing encloses the end windings **430**. The housing **408** may further include coolant channels **434** in a lower section **436**. Coolant may flow from the channels **434** into a sump **438** as indicated via arrows **440**. In this way, coolant that has been used to cool the end windings returns to the sump. In the illustrated example, the coolant channels **434** extend axially outward through the housing. However, in other examples, the coolant channels **434** may extend both axially and radially downward to increase the coolant flowrate through the channels. The pump **404** may include a pick-up **442** in the sump **438** that allows a coolant loop to be formed in the machine. From the pump, coolant flows back to the lines **406**.

(46) FIG. 4 further shows a rotor **444** with a rotor shaft **446** and a rotor core **448** that has balancing plates **450** coupled to opposing sides of the rotor core. The balancing plates **450** may be contoured to reduce rotational imbalances in the machine.

(47) FIG. 5 shows a detailed view of the end windings **430** and one of the spray rings **412** included in the electric machine **400**, shown in FIG. 4. The first section **418**, the second section **422** and the third section **424** of the spray ring are again shown which form “C” type shape. The third section **424** may shields coolant spray from the centrifugal airflow originating from the rotor and, therefore, increase the effectiveness of the cooling methodology in the electric machine. A surface **500** at the inner diameter of the spray ring may have radius **501** that is greater than or equal to an outer radius **502** of the air gap **504**. In this way, the amount of coolant entering the air gap is reduced, thereby increasing machine efficiency due to the reduction in drag losses. The third section **424** of the spray ring may taper in a radially outward direction to allow the coolant to achieve a desired flow pattern which increases end winding cooling while reducing the amount of

coolant that enters the air gap **504**. Further, the second section **422** may have a greater axial length than the third section **424** to achieve a desired coolant flow around and through the end windings which increased coolant penetration into the end windings.

(48) Further, the first section **418** of the spray ring may have a greater thickness than the second section **422** to allow the spray ring to generate a desired coolant flow pattern. However, other spray ring profiles have been envisioned.

(49) Further, the coolant compartment **410** extends down the length of an outer surface **510** of the second section **422** of the spray ring **412**, in the illustrated example. However, in other examples, the boundary of the coolant compartment may be decreased such that the compartment is arranged in a region closer to the spray channel **428**, in other examples. FIG. 5 further shows a layer of adhesive **512** that may be used to bond the spray ring **412** to the stator core **414**. However, other techniques for attachment between the spray ring and the stator core may be used, in other examples.

(50) FIG. 6 shows another example of a spray ring **600**. The spray ring **600** is again illustrated directly coupled to an axial end face **602** of a stator core **604**. The spray ring **600** again includes a first section **606** and a second section **608** that form an “L” shape. The “L” shaped spray ring enables a greater amount of coolant to flow through the end windings **610**.

(51) The spray rings with the features described herein enable end winding cooling to be effectively increased using an assembly that is efficient to manufacture and install. The appeal of the machine is therefore increased.

(52) FIGS. 1-6 provide for the following method for operation of an electric machine. The method includes spraying a coolant from a spray ring toward stator end windings. The method may further include, prior to the spraying step, flowing coolant through a passage that traverses the stator core. In this example the electric machine includes a cooling assembly with a spray ring that is directly coupled to an axial face of the stator core. Further, the spray ring is configured to spray a coolant at the end windings and includes a first section that extends down the axial face of the stator core and a second section that extends axially away from the stator core.

(53) The technical effect of the method for electric machine operation is to increase machine efficiency by transferring greater amount of heat away from the end windings, when compared to other machine designs.

(54) FIG. 3 is drawn approximately to scale. Although other relative component dimensions may be used, in other embodiments.

(55) FIGS. 1-6 show example configurations with relative positioning of the various components. If shown directly contacting each other, or directly coupled, then such elements may be referred to as directly contacting or directly coupled, respectively, at least in one example. Similarly, elements shown contiguous or adjacent to one another may be contiguous or adjacent to each other, respectively, at least in one example. As an example, components laying in face-sharing contact with each other may be referred to as in face-sharing contact. As another example, elements positioned apart from each other with only a space there-between and no other components may be referred to as such, in at least one example. As yet another example, elements shown above/below one another, at opposite sides to one another, or to the left/right of one another may be referred to as such, relative to one another. Further, as shown in the figures, a topmost element or point of element may be referred to as a “top” of the component and a bottommost element or point of the element may be referred to as a “bottom” of the component, in at least one example. As used herein, top/bottom, upper/lower, above/below, may be relative to a vertical axis of the figures and used to describe positioning of elements of the figures relative to one another. As such, elements shown above other elements are positioned vertically above the other elements, in one example. As yet another example, shapes of the elements depicted within the figures may be referred to as having those shapes (e.g., such as being circular, straight, planar, curved, rounded, chamfered, angled, or the like). Additionally, elements co-axial with one another may be referred to as such, in

one example. Further, elements shown intersecting one another may be referred to as intersecting elements or intersecting one another, in at least one example. Further still, an element shown within another element or shown outside of another element may be referred to as such, in one example. In other examples, elements offset from one another may be referred to as such.

(56) The invention will be further described in the following paragraphs. In one aspect, an electric machine is provided that comprises a stator including a plurality of end windings and a stator core; and a cooling assembly including a spray ring that is directly coupled to an axial face of the stator core, wherein the spray ring is configured to spray a coolant at the plurality of end windings and includes: a first section that extends down the axial face of the stator core; and a second section that extends axially away from the stator core.

(57) In another aspect, a method for operating a cooling assembly in an electric machine is provided that comprises spraying a coolant from a spray ring toward a plurality of stator end windings; wherein the electric machine includes: a cooling assembly that includes the spray ring that is directly coupled to an axial face of the stator core, wherein the spray ring is configured to spray a coolant at the plurality of end windings and includes: a first section that extends down the axial face of the stator core; and a second section that extends axially away from the stator core. The method may further comprise, in one example, flowing coolant through a passage that traverses the stator core.

(58) In yet another example, an electric motor is provided that comprises a stator including a plurality of hairpin end windings and a stator core; and a cooling assembly including a spray ring that is directly coupled to an axial face of the stator core, wherein the spray ring is configured to spray oil at the plurality of end windings and includes: a first section that extends down the axial face of the stator core; and a second section that extends axially away from the stator core.

(59) In any of the aspects or combinations of the aspects, the coolant may include oil.

(60) In any of the aspects or combinations of the aspects, the spray ring may be directly coupled to a stack of stator laminations in the stator core.

(61) In any of the aspects or combinations of the aspects, the spray ring may be bonded and impregnated to the stack of stator laminations.

(62) In any of the aspects or combinations of the aspects, the plurality of stator end windings may be hairpin end windings.

(63) In any of the aspects or combinations of the aspects, the spray ring may include a coolant passage that extends through the second section.

(64) In any of the aspects or combinations of the aspects, the electric machine may be a multi-phase electric machine.

(65) In any of the aspects or combinations of the aspects, the electric machine may be a traction motor in an electric drive of a vehicle.

(66) In any of the aspects or combinations of the aspects, the cooling assembly may include a sump configured to collect the coolant.

(67) In any of the aspects or combinations of the aspects, the electric machine may further comprise a pump configured to receive coolant from the sump, wherein the pump is incorporated within the electric machine.

(68) In any of the aspects or combinations of the aspects, the spray ring may be adhesively bonded to the stack of stator laminations.

(69) In any of the aspects or combinations of the aspects, a portion of the second section of the spray ring may be incorporated into a recess of a housing.

(70) In any of the aspects or combinations of the aspects, an interface between the spray ring and the stator core may not include a seal.

(71) In any of the aspects or combinations of the aspects, the spray ring may include a third section that extends axially away from the stator core at an inner diameter of the spray ring.

(72) In any of the aspects or combinations of the aspects, the third section may be positioned at an

inner diameter of the spray ring.

(73) In another representation, a stator assembly in an electric motor is provided that comprises a spray ring directly bonded to a peripheral surface of an outer stator lamination in a stack of stator laminations, wherein the spray ring is configured to spray oil towards hairpin end windings, wherein the spray ring is L-shaped and includes a section that at least partially circumferentially extends around the hairpin end windings.

(74) As used herein, the term “approximately” is construed to mean plus or minus five percent of the range, unless otherwise specified.

(75) Note that the example control and estimation routines included herein can be used with various motor configurations. The control methods and routines disclosed herein may be stored as executable instructions in non-transitory memory and may be carried out by the control system including the controller in combination with the various sensors, actuators, and other system hardware in combination with the electronic controller. As such, the described actions, operations, and/or functions may graphically represent code to be programmed into non-transitory memory of the computer readable storage medium in the vehicle and/or driveline control system. One or more of the illustrated actions, operations and/or functions may be repeatedly performed depending on the particular strategy being used. One or more of the method steps described herein may be omitted if desired.

(76) While various embodiments have been described above, it should be understood that they have been presented by way of example, and not limitation. It will be apparent to persons skilled in the relevant arts that the disclosed subject matter may be embodied in other specific forms without departing from the spirit of the subject matter. The embodiments described above are therefore to be considered in all respects as illustrative, not restrictive. As such, the configurations and routines disclosed herein are exemplary in nature, and that these specific examples are not to be considered in a limiting sense, because numerous variations are possible. The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various systems and configurations, and other features, functions, and/or properties disclosed herein.

(77) The following claims particularly point out certain combinations and sub-combinations regarded as novel and non-obvious. These claims may refer to “an” element or “a first” element or the equivalent thereof. Such claims should be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements. Other combinations and sub-combinations of the disclosed features, functions, elements, and/or properties may be claimed through amendment of the present claims or through presentation of new claims in this or a related application. Such claims, whether broader, narrower, equal, or different in scope to the original claims, also are regarded as included within the subject matter of the present disclosure.

Claims

1. An electric machine, comprising: a stator including a plurality of hairpin end windings and a stator core; and a cooling assembly including a spray ring that is directly coupled to an axial face of the stator core, wherein the spray ring is configured to spray a coolant at the plurality of hairpin end windings and includes: a first section that extends down the axial face of the stator core; and a second section that extends axially away from the stator core; wherein the second section includes an interior surface with a step that alters coolant flow; wherein the step is positioned axially inward from a plurality of tips of the plurality of hairpin end windings; wherein the spray ring is directly adhesively bonded to a stack of stator laminations; wherein the second section is mated with a recess in a housing; and wherein the housing includes a coolant compartment that is positioned radially outward from the step.

2. The electric machine of claim 1, wherein the spray ring includes a third section that extends axially away from the stator core.

3. The electric machine of claim 2, wherein the third section is positioned at an inner diameter of the spray ring.
 4. The electric machine of claim 1, wherein the spray ring includes a coolant opening that extends through the second section.
 5. The electric machine of claim 1, wherein the electric machine is a multi-phase electric machine.
 6. The electric machine of claim 1, wherein the coolant is oil.
 7. The electric machine of claim 1, wherein the electric machine is a traction motor in an electric drive of a vehicle.
 8. The electric machine of claim 1, wherein the cooling assembly includes a sump configured to collect the coolant.
 9. The electric machine of claim 8, further comprising a pump configured to receive coolant from the sump, wherein the pump is incorporated within the electric machine.
 10. A method for operating a cooling assembly in an electric machine, comprising: spraying a coolant from a spray ring toward a plurality of stator hairpin end windings; wherein the electric machine includes: a cooling assembly that includes the spray ring that is directly coupled to an axial face of a stator core, wherein the spray ring includes: a first section that extends down the axial face of the stator core; and a second section that extends axially away from the stator core; wherein the second section includes an interior surface with a step that alters coolant flow; wherein the step is positioned axially inward from a plurality of tips of the plurality of hairpin end windings; wherein the spray ring is directly adhesively bonded to a stack of stator laminations; wherein the second section is mated with a recess in a housing; and wherein the housing includes a coolant compartment that is positioned radially outward from the step.
 11. The method of claim 10, further comprising flowing coolant through a passage that traverses the stator core.
 12. An electric motor, comprising: a stator including a plurality of hairpin end windings and a stator core; and a cooling assembly including a spray ring that is directly coupled to an axial face of the stator core, wherein the spray ring is configured to spray oil at the plurality of end windings and includes: a first section that extends down the axial face of the stator core; and a second section that extends axially away from the stator core; wherein the second section includes an interior surface with a step that alters coolant flow; wherein the step is positioned axially inward from a plurality of tips of the plurality of hairpin end windings; wherein the spray ring is directly adhesively bonded to the stator core; wherein the second section is mated with a recess in a housing; and wherein the housing includes a coolant compartment that is positioned radially outward from the step.
 13. The electric motor of claim 12, wherein the spray ring includes a third section that extends axially away from the stator core at an inner diameter of the spray ring.
 14. The electric motor of claim 12, wherein the spray ring is adhesively bonded to the stator core.
 15. The electric motor of claim 12, wherein a portion of the second section of the spray ring is incorporated into a recess of a housing.
 16. The electric motor of claim 12, wherein an interface between the spray ring and the stator core does not include an O-ring seal.
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